Ch 07: Tuples and Plotting Sequences with Matplotlib Library

Application Program Development

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Tuples

- **Tuple**: an immutable sequence
  - Very similar to a list
  - Once it is created it cannot be changed
  - Format: 't = (item1, item2)
  - Tuples support operations as lists
    - Subscript indexing for retrieving elements
    - Methods such as `index()`
    - Built in functions such as `len()`, `min()`, `max()`
    - slicing expressions
    - The `in` `+` and `*` operators
  - Tuples do not support
    - `append()`
    - `remove()`
    - `insert()`
    - `reverse()`
    - `sort()`

Tuple

A Tuple is an immutable sequence, which means that its contents cannot be changed.

Advantages of tuples over lists:

- Processing tuples is faster than processing lists
- Tuples are safe
- Some operations in Python require use of tuples

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Plotting Data with `matplotlib` Library

- The `matplotlib` package is a library for creating two-dimensional charts and graphs
- It is **not** part of the standard Python library, so you will have to install it separately (e.g. `pip install`) after you install Python
- The `matplotlib` package contains a (sub) module named `pyplot` that you will need to import
- Conventionally the following is used to import the `pyplot` module into the `plt` name space:

```python
import matplotlib.pyplot as plt
```

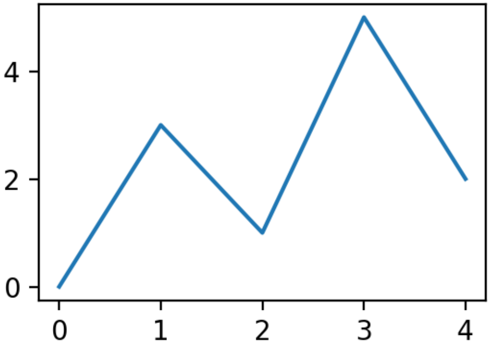
Consult the Matplotlib documentation for reference and more plot types (Matplotlib, 2026).

Plotting a Line Graph with the `plot()` Function

- Use the `plot()` function to create a line graph that connects a series of points with straight lines
  - Each point has an `(x, y)` coordinate
  - Notice 2 separate sequences of same size, one for `x` and one for `y` coordinates

```python
import matplotlib.pyplot as plt

x = [0, 1, 2, 3, 4]
y = [0, 3, 1, 5, 2]
plt.plot(x, y)
```

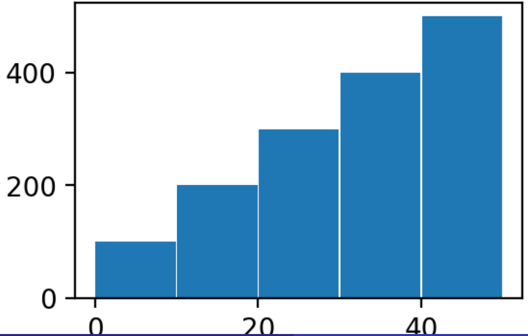


Plotting a Bar Chart

- Use the `bar()` function in the `matplotlib.pyplot` module to create bar charts
- The function needs two sequences
  1. one with the x coordinates of each bar's center
  2. and another with the heights of each bar along the y axis

```python
import matplotlib.pyplot as plt

centers = [5, 15, 25, 35, 45]
heights = [100, 200, 300, 400, 500]
bar_width = 9.8
plt.bar(centers, heights, bar_width)
```

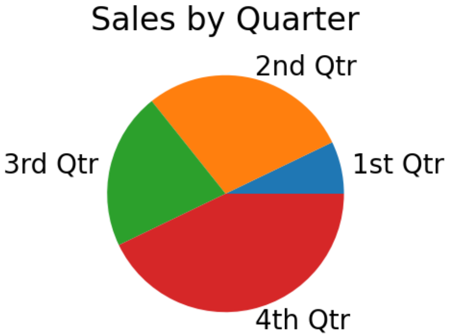


Plotting a Pie Chart

- Use the `pie()` function in the `matplotlib.pyplot` module to create a pie chart
- When you call the `pie()` function, you pass a list of values
  - The sum of the values will be used as the value of the whole
  - Each element in the list will become a slice in the pie chart
  - The size of a slice represents the element's value as a percentage of the whole
- The `pie()` function has a `labels` parameter that you can use to display labels for slices
  - A list of strings for the desired labels

```
import matplotlib.pyplot as plt

sales = [100, 400, 300, 600]
labels = ['1st Qtr', '2nd Qtr', '3rd Qtr', '4th
plt.pie(sales, labels=labels)
plt.title('Sales by Quarter')
```



Summary

- A tuple is an immutable sequence, similar to a list, but it cannot be changed once created
- Has several advantages over a list, and must be used in some situations
- You can use the `matplotlib.pyplot` library module to perform basic plotting
    - Basically plot functions take sequences (lists or tuples) to specify (x,y) coordinates, bar location and height, or other plot elements
    - Can plot line graphs, bar charts, pie charts, and many more not shown

References

Matplotlib. (2026). *Matplotlib module documentation*. Retrieved August 13, 2026, from https://matplotlib.org/stable/index.html